Yang Mills Existence And Mass Gap.hypotheses Evaluated

Hypothesis Evaluation Report

Stats

- Hypotheses evaluated: 44
- Support threshold: 0.62
- Refute threshold: 0.62
- Margin: 0.08
- Force resolution: False
- Coherence signals: 44
- Global coherence baseline: 0.320043

Verdict Counts

- inconclusive: 32
- supported: 12

Results

hypothesis_id	verdict	next_status	support	refute	confidence
phyp.yang-mills-existence-and-mass-gap.021	inconclusive	needs-testing	0.4708	0.0000	0.3587
phyp.yang-mills-existence-and-mass-gap.022	inconclusive	needs-testing	0.4302	0.0000	0.3199
phyp.yang-mills-existence-and-mass-gap.023	inconclusive	needs-testing	0.4551	0.0046	0.3425
phyp.yang-mills-existence-and-mass-gap.001	inconclusive	needs-testing	0.4709	0.0540	0.3474
phyp.yang-mills-existence-and-mass-gap.024	inconclusive	needs-testing	0.5375	0.0047	0.4244
phyp.yang-mills-existence-and-mass-gap.025	inconclusive	needs-testing	0.4495	0.0050	0.3371
phyp.yang-mills-existence-and-mass-gap.002	inconclusive	needs-testing	0.4761	0.0792	0.3469
phyp.yang-mills-existence-and-mass-gap.026	inconclusive	needs-testing	0.4662	0.0000	0.3542
phyp.yang-mills-existence-and-mass-gap.003	inconclusive	needs-testing	0.4688	0.0180	0.3530
phyp.yang-mills-existence-and-mass-gap.004	inconclusive	needs-testing	0.4676	0.0540	0.3442
phyp.yang-mills-existence-and-mass-gap.005	inconclusive	needs-testing	0.5054	0.0900	0.3724
phyp.yang-mills-existence-and-mass-gap.006	inconclusive	needs-testing	0.4593	0.0540	0.3364
phyp.yang-mills-existence-and-mass-gap.033	supported	supported	0.7364	0.0528	0.6315

hypothesis_id	verdict	next_status	support	refute	confidence
phyp.yang-mills-existence-and-mass-gap.027	inconclusive	needs-testing	0.4427	0.0057	0.3306
phyp.yang-mills-existence-and-mass-gap.007	inconclusive	needs-testing	0.4478	0.0000	0.3365
phyp.yang-mills-existence-and-mass-gap.008	inconclusive	needs-testing	0.4878	0.1196	0.3491
phyp.yang-mills-existence-and-mass-gap.034	supported	supported	0.6620	0.1269	0.5235
phyp.yang-mills-existence-and-mass-gap.009	inconclusive	needs-testing	0.4923	0.0360	0.3718
phyp.yang-mills-existence-and-mass-gap.036	supported	supported	0.7302	0.0576	0.6226
phyp.yang-mills-existence-and-mass-gap.035	supported	supported	0.6951	0.0312	0.5900
phyp.yang-mills-existence-and-mass-gap.037	supported	supported	0.7594	0.0792	0.6501
phyp.yang-mills-existence-and-mass-gap.010	inconclusive	needs-testing	0.4544	0.0809	0.3263
phyp.yang-mills-existence-and-mass-gap.038	supported	supported	0.6799	0.0576	0.5644
phyp.yang-mills-existence-and-mass-gap.011	inconclusive	needs-testing	0.4443	0.0360	0.3260
phyp.yang-mills-existence-and-mass-gap.012	inconclusive	needs-testing	0.4104	0.0360	0.2949
phyp.yang-mills-existence-and-mass-gap.013	inconclusive	needs-testing	0.4667	0.0000	0.3547
phyp.yang-mills-existence-and-mass-gap.014	inconclusive	needs-testing	0.4401	0.0900	0.3114
phyp.yang-mills-existence-and-mass-gap.015	inconclusive	needs-testing	0.4097	0.0798	0.2862
phyp.yang-mills-existence-and-mass-gap.039	supported	supported	0.6309	0.0000	0.5261
phyp.yang-mills-existence-and-mass-gap.028	inconclusive	needs-testing	0.4040	0.0069	0.2944
phyp.yang-mills-existence-and-mass-gap.016	inconclusive	needs-testing	0.4484	0.0957	0.3177
phyp.yang-mills-existence-and-mass-gap.040	supported	supported	0.6317	0.1572	0.4823
phyp.yang-mills-existence-and-mass-gap.017	inconclusive	needs-testing	0.4245	0.1172	0.2922
phyp.yang-mills-existence-and-mass-gap.018	inconclusive	needs-testing	0.4486	0.0360	0.3300
phyp.yang-mills-existence-and-mass-gap.041	supported	supported	0.7829	0.0216	0.6988
phyp.yang-mills-existence-and-mass-gap.019	inconclusive	needs-testing	0.4304	0.0360	0.3131
phyp.yang-mills-existence-and-mass-gap.020	inconclusive	needs-testing	0.4995	0.0825	0.3684
phyp.yang-mills-existence-and-mass-gap.042	supported	supported	0.6260	0.1305	0.4839
phyp.yang-mills-existence-and-mass-gap.029	inconclusive	needs-testing	0.4762	0.0074	0.3623
phyp.yang-mills-existence-and-mass-gap.043	supported	supported	0.7011	0.0264	0.5985
phyp.yang-mills-existence-and-mass-gap.044	supported	supported	0.6296	0.0264	0.5171
phyp.yang-mills-existence-and-mass-gap.030	inconclusive	needs-testing	0.5020	0.0000	0.3895
phyp.yang-mills-existence-and-mass-gap.031	inconclusive	needs-testing	0.2888	0.0068	0.1955
phyp.yang-mills-existence-and-mass-gap.032	inconclusive	needs-testing	0.3327	0.0113	0.2312

Mathematical Readout

phyp.yang-mills-existence-and-mass-gap.021 [intervention-focus]

- Verdict: inconclusive (support=0.4708, refute=0.0000, confidence=0.3587)
- Alignment signals: antecedent=0.0000, consequence=0.0000, invariant=0.0000, barrier=0.0000
- Key reasons: high priority inherited from connector hypotheses; high downstream impact relative to intervention set

phyp.yang-mills-existence-and-mass-gap.022 [**intervention-focus**]

- Verdict: inconclusive (support=0.4302, refute=0.0000, confidence=0.3199)
- Alignment signals: antecedent=0.0000, consequence=0.0000, invariant=0.0000, barrier=0.0000
- Key reasons: high priority inherited from connector hypotheses

phyp.yang-mills-existence-and-mass-gap.023 [**intervention-focus**]

- Verdict: inconclusive (support=0.4551, refute=0.0046, confidence=0.3425)
- Alignment signals: antecedent=0.0000, consequence=0.0000, invariant=0.0000, barrier=0.0000
- Key reasons: high priority inherited from connector hypotheses; strong cross-component potential

phyp.yang-mills-existence-and-mass-gap.001 [**bridge-link**]

- Verdict: inconclusive (support=0.4709, refute=0.0540, confidence=0.3474)
- Antecedent schema: reflection-positivity-and-transfer-matrix
- Target consequence: positive-self-adjoint-Hamiltonian
- Preserved invariant: OS-to-Wightman mass-gap transfer in Hamiltonian-mass-gap
- Barrier scope: continuum-limit
- Alignment signals: antecedent=0.3869, consequence=0.0000, invariant=0.5114, barrier=0.0000
- Key reasons: structural signature overlap between support and target; symbol-role alignment indicates semantic compatibility; theorem-family embedding similarity indicates structural proximity; high connector analysis score for support->target pair

phyp.yang-mills-existence-and-mass-gap.024 [**intervention-focus**]

- Verdict: inconclusive (support=0.5375, refute=0.0047, confidence=0.4244)
- Alignment signals: antecedent=0.0000, consequence=0.0000, invariant=0.0000, barrier=0.0000
- Key reasons: high priority inherited from connector hypotheses; strong cross-component potential

phyp.yang-mills-existence-and-mass-gap.025 [**intervention-focus**]

- Verdict: inconclusive (support=0.4495, refute=0.0050, confidence=0.3371)
- Alignment signals: antecedent=0.0000, consequence=0.0000, invariant=0.0000, barrier=0.0000
- Key reasons: high priority inherited from connector hypotheses; strong cross-component potential

phyp.yang-mills-existence-and-mass-gap.002 [bridge-link]

- Verdict: inconclusive (support=0.4761, refute=0.0792, confidence=0.3469)
- Antecedent schema: unconditional
- Target consequence: moduli-space-control
- Preserved invariant: mass-gap witness transfer in compact-simple-gauge-group
- Barrier scope: none
- Alignment signals: antecedent=0.0000, consequence=0.2536, invariant=0.3859, barrier=0.0000
- Key reasons: structural signature overlap between support and target; symbol-role alignment indicates semantic compatibility; theorem-family embedding similarity indicates structural proximity; high connector analysis score for support->target pair

phyp.yang-mills-existence-and-mass-gap.026 [intervention-focus]

- Verdict: inconclusive (support=0.4662, refute=0.0000, confidence=0.3542)
- Alignment signals: antecedent=0.0000, consequence=0.0000, invariant=0.0000, barrier=0.0000
- Key reasons: high priority inherited from connector hypotheses; high downstream impact relative to intervention set

phyp.yang-mills-existence-and-mass-gap.003 [bridge-link]

- Verdict: inconclusive (support=0.4688, refute=0.0180, confidence=0.3530)
- Antecedent schema: reflection-positivity-and-transfer-matrix
- Target consequence: positive-self-adjoint-Hamiltonian
- Preserved invariant: correlator-decay mass-gap witness in Schwinger-function-decay
- Barrier scope: continuum-limit
- Alignment signals: antecedent=0.3869, consequence=0.3815, invariant=0.4388, barrier=0.0000
- Key reasons: structural signature overlap between support and target; symbol-role alignment indicates semantic compatibility; theorem-family embedding similarity indicates structural proximity; high connector analysis score for support->target pair

phyp.yang-mills-existence-and-mass-gap.004 [bridge-link]

- Verdict: inconclusive (support=0.4676, refute=0.0540, confidence=0.3442)
- Antecedent schema: constructive-existence
- Target consequence: rigorous-Yang-Mills-construction
- Preserved invariant: OS-to-Wightman mass-gap transfer in Hamiltonian-mass-gap
- Barrier scope: ultraviolet-control

- Alignment signals: antecedent=0.0000, consequence=0.1910, invariant=0.5109, barrier=0.0000
- Key reasons: symbol-role alignment indicates semantic compatibility; theorem-family embedding similarity indicates structural proximity; high connector analysis score for support->target pair; target consequence schema grounded in target statement structure: rigorous-Yang-Mills-construction

phyp.yang-mills-existence-and-mass-gap.005 [bridge-link]

- Verdict: inconclusive (support=0.5054, refute=0.0900, confidence=0.3724)
- Antecedent schema: constructive-existence
- Target consequence: positive-self-adjoint-Hamiltonian
- Preserved invariant: OS-to-Wightman mass-gap transfer in Hamiltonian-mass-gap
- Barrier scope: continuum-limit
- Alignment signals: antecedent=0.0000, consequence=0.0000, invariant=0.5119, barrier=0.0000
- Key reasons: structural signature overlap between support and target; symbol-role alignment indicates semantic compatibility; theorem-family embedding similarity indicates structural proximity; high connector analysis score for support->target pair

phyp.yang-mills-existence-and-mass-gap.006 [bridge-link]

- Verdict: inconclusive (support=0.4593, refute=0.0540, confidence=0.3364)
- Antecedent schema: reflection-positivity-and-transfer-matrix
- Target consequence: positive-self-adjoint-Hamiltonian
- Preserved invariant: transfer-matrix spectral witness in reflection-positive-transfer-matrix
- Barrier scope: continuum-limit
- Alignment signals: antecedent=0.0000, consequence=0.3815, invariant=0.3827, barrier=0.0000
- Key reasons: structural signature overlap between support and target; symbol-role alignment indicates semantic compatibility; theorem-family embedding similarity indicates structural proximity; high connector analysis score for support->target pair